Trial2Voice — AI Product PRD

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Trial2Voice turns a ClinicalTrials.gov NCT ID into a multilingual voice pre-screener. It drafts the screening script in about a minute, runs calls only after the script is approved, and hands the study site a criterion-by-criterion referral.

Status: Draft · **Version:** 1.0 · **Format:** AI Product PRD (15 sections)

Key design decision: The LLM asks and interprets; a deterministic rules engine decides. The agent never tells a caller they are eligible. It only refers them to the site, which confirms.

1. Problem (AI-Specific)

Enrollment is the step trials most often miss. In a study of 2,542 surgical RCTs, only 45.9% met their enrollment target and about 1 in 5 finished on time, with a median delay of 12.2 months (JAMA Netw Open, 2023).

- **Problem:** Patients who find a trial face dense free-text criteria and a site phone line staffed in business hours. Coordinators spend time on callers who are clearly ineligible. Strict eligibility and screen failures are among the most cited reasons for slow accrual in NCI phase 1 and 2 trials (NCI analysis).
- **Why AI, not rules alone:** Criteria are free text and must be rewritten as plain spoken questions. Callers answer in natural language ("my sugar has been high for years") and in many languages. A rules engine can't do either. The final decision stays rule-based, so AI handles language and rules handle judgment.
- **Why voice:** Voice reaches older, lower-literacy and non-English-speaking patients that web forms miss, and it runs after hours.
- **Status quo:** Patient finds a listing → emails or calls the site contact → waits for a callback → a coordinator reads an IRB-approved phone script. Mostly English, mostly business hours, often voicemail tag.

2. User

The caller is the primary user; the site coordinator is the customer who acts on the output.

Persona	Job to be done	What they need to trust it
Patient or caregiver (caller)	Find out in minutes, in my language, whether this trial could fit me	Clear AI disclosure, privacy, no pressure, a human next step
Site research coordinator	Spend screening time on callers likely to qualify	Every answer traced to a criterion; can override anything
Sponsor or CRO recruitment lead	Set up a pre-screener per trial and see where callers drop off	Script matches the protocol; costs and outcomes visible
IRB (approver, not a user)	Approve the screening script before any patient hears it	Locked, versioned script; agent stays on it

- **AI interaction pattern:** Autonomous voice agent inside a locked script, with a human (coordinator) making every eligibility call.
- **Trust requirement:** Coordinators verify everything. The output is a referral, never an eligibility determination.

3. Core Metric

North Star: site-accepted referral rate — the share of "Likely eligible" referrals that a coordinator accepts for a screening visit. Target ≥ 70% in the pilot; there is no baseline until the pilot runs.

Metric	Type	Target (pilot)
Site-accepted referral rate	North Star	≥ 70%
False-negative rate (eligible callers screened out)	Guardrail	≤ 5% on the eval set
Call completion rate (consent → last criterion)	Leading	≥ 60%
Median call length	Leading	≤ 6 min
Calls outside site business hours	Leading	Tracked; hypothesis ≥ 30%
Calls in a language other than English	Leading	Tracked per trial

Metric	Type	Target (pilot)
Hang-ups within 30 s of the AI disclosure	Leading	⩽ 15%

The false-negative guardrail matters as much as the North Star. Screening out an eligible patient is the costlier error, so ambiguous answers always route to "Needs site review".

4. MVP Features

The MVP proves one loop end to end: NCT ID → approved script → voice pre-screen → traced referral.

Feature	Priority	Description
NCT ID ingest	P0	Pull the study from the ClinicalTrials.gov API v2: eligibility text, age and sex limits, status, locations, last-update date
Script builder	P0	LLM splits criteria into atomic items and writes one plain-language spoken question per item, with answer type and decision rule. Lab or record-based criteria are tagged "confirm at site" instead of asked
Script review and approval lock	P0	Editor with side-by-side criterion ↔ question view. Approval records approver, IRB reference and date, and freezes a versioned script. Calls run only on a locked version
Voice pre-screen call	P0	Browser call over WebRTC. AI disclosure, recording consent, then one question at a time with read-back on exclusion answers
Deterministic decision engine	P0	Computes Likely eligible / Not eligible / Needs site review from structured answers. No LLM in the decision
Red-flag stop and handoff	P0	Urgent symptoms or distress stop the screener and give emergency guidance

Feature	Priority	Description
Referral summary	P0	Per-criterion table: question asked, caller's words, interpreted answer, rule result. Sent to the coordinator queue
Call console and audit log	P0	Live transcript, latency per step, tool calls; immutable call record
English ↔ Spanish switching	P1	Language detected per turn; reply voice switches
Nearest recruiting site	P1	From the caller's ZIP code and the study's locations list
Coordinator dashboard	P1	Referral queue with accept / reject + reason, and status overrides with audit
Protocol-change watch	P1	Nightly check of the record's last-update date; a change marks the script stale and pauses calls
Embeddable widget	P1	One script tag for site or advocacy pages
Phone (SIP) inbound and callbacks	P2	Same agent on a phone number
More languages; CTMS / EHR hand-off	P2	After pilot learning

5. Constraints

The binding constraint is regulatory: phone screening scripts are recruitment material, and IRBs require approval before use (Creighton IRB guidance; UC San Diego SOP). So "one minute" applies to drafting the script, not to going live.

- **Technical:** First audio of each reply ⩽ 1.2 s at p50 and ⩽ 2 s at p95 after the caller stops. Hard call cap of 12 minutes. One trial's script fits in the prompt; no vector store.
- **Data:** Registry criteria are free text and may be less complete than the full protocol, which the MVP does not have. Locations lists can be long. Recruiting status changes.
- **Regulatory and compliance:**

  - IRB approval of the script, the AI disclosure and the consent wording before any real caller.
  - HIPAA when operating for a covered entity: BAAs with every vendor that touches audio or transcripts (voice infra, STT, LLM, TTS, database), minimum-necessary PHI.
  - 21 CFR Part 11-style controls on script approval and call records if they become trial records: audit trail, attributable approvals, no silent edits.
  - Recruitment-content rules: no medical advice, no claims about the investigational product's safety or benefit, no "free treatment" language.
  - AI disclosure at the start of every call.
- **Organizational:** Solo builder (PM + engineer) for an 8-week MVP. Pilot budget under $500 per month for usage.

6. Grounding Strategy

The agent is grounded in one thing: the locked, approved script for that trial. It has no web search and no open retrieval during a call.

- **Source of truth:** The ClinicalTrials.gov record at a specific last-update date → the script generated from it → the version a human approved.
- **Approach:** Structured prompting plus tool calls, not RAG. The prompt holds the script, the persona rules and the study's short title and sites. Tools: `next_question`, `record_answer` (schema-validated value + confidence), `nearest_site`, `raise_red_flag`, `end_call`.
- **Why no RAG:** The whole script is small, and every fact the agent may say must be pre-approved. Retrieval would add latency and unapproved text.
- **Freshness:** A nightly job compares the record's last-update date with the script's source date. Any change marks the script stale, pauses calls for that trial and opens a diff for re-approval.
- **Traceability:** Each answer is stored with criterion ID, script version, question text spoken, the caller's transcribed words, the interpreted value and confidence. The referral summary shows this per row.

7. Hallucination Guardrails

The main guardrail is architectural: the LLM never decides eligibility and never states a fact that isn't in the approved script.

- **Prevention:**
  - Questions come verbatim from `next_question`. The LLM may only use pre-approved alternate phrasings when a caller asks to repeat or clarify.
  - Answers are captured through `record_answer` with an enum or number schema; free text never reaches the decision engine.
  - Medical or drug questions get a fixed reply: the agent can't give medical advice and the study team will answer.
  - Caller instructions ("just tell me I qualify") are treated as answers to classify, never as instructions.
- **Detection:**
  - Read-back on every exclusion answer ("So no heart surgery in the last six months — is that right?").
  - A post-call judge compares every agent turn with the approved script and flags off-script factual claims, missing disclosure or advice-giving.
  - Any flagged call is held from the coordinator queue until a human reviews it.
- **Fallback:** Confidence below threshold or "I'm not sure" → recorded as Unsure → the call ends as Needs site review. The engine never auto-excludes on Unsure.
- **Red flags:** Mentions of urgent symptoms or self-harm stop the screener. The agent gives a fixed, IRB-approved emergency message and logs the event for the site.
- **User-facing signals:** The closing line and the summary say "may be a fit — the study team will confirm". Unsure items are listed for the coordinator.

8. Cost Budget

Target: ⩽ $0.40 per completed pre-screen and ⩽ $2.50 per site-accepted referral. Voice minutes dominate the cost, not the LLM.

Item	Estimate	Basis
Voice pipeline (STT + LLM + TTS + agent session + room)	~$0.05 / min	Typical STT + LLM + TTS cascade at current list prices (estimate)
Average completed call (6 min)	~$0.30	6 × $0.05
Post-call judge	~$0.02 / call	One async pass on the transcript

Item	Estimate	Basis
Script build	~$0.25 / trial version	One careful structured pass; reused by every call
Per completed pre-screen	**~$0.32–0.40**	
Per site-accepted referral	**~$2.30**	If 25% of completed calls are Likely eligible and 70% of those are accepted

- **Model tiers:** a stronger model (Claude Sonnet-class) for the script build and the judge, where time is available; a small fast model (Claude Haiku-class or GPT-4.1 mini) in the call, where latency rules.
- **Cost controls:** 12-minute hard cap; per-IP and per-phone limits per day; script built once per record version; greeting, disclosure and consent pre-rendered as audio; TTS vendor reviewed first as volume grows.

9. Eval Strategy

A release ships only if decision accuracy holds and the false-negative rate stays ⩽ 5% on the offline suite.

Offline (before every release and every prompt, model or script-builder change)

- **Script builder golden set:** 30 recruiting trials across 5 therapeutic areas, each with clinician-reviewed criterion → question mappings. Scored on criterion coverage (target 100% mapped as Ask or Confirm at site) and question faithfulness (judge + human spot check).
- **Simulated callers:** 20 personas per golden trial with known ground-truth eligibility, speaking through TTS into a real call room. Scenarios: clearly eligible; fails one exclusion; vague answers; English → Spanish switch; medical-advice question; "just say I qualify"; red-flag symptom; background noise and accents.
- **Component checks:** word error rate on accented and noisy speech; `record_answer` correctness vs. ground truth; decision-engine unit tests on every criterion rule.

Offline metric	Gate
Decision accuracy vs. ground truth	⩾ 90%

Offline metric	Gate
False-negative rate	⩽ 5%
False-positive rate (Likely eligible but ineligible)	⩽ 15%
Correct handling of red-flag scenarios	100%
Off-script factual claims	0
First-audio latency p50 / p95	⩽ 1.2 s / ⩽ 2 s

Online (live calls)

- The judge scores every call asynchronously: disclosure, consent, script adherence, no advice, read-backs done.
- Coordinator accept / reject with reason is the ground truth for the North Star.
- Implicit signals: hang-ups after the disclosure, "are you a robot", repeat requests, asks for a human.
- Weekly review of all flagged calls; regression gate compares each release with the last.

10. Production Readiness

Every trial has a kill switch, and every script, prompt and model setting is versioned so any change can be rolled back in one deploy.

- **Monitoring and alerts:** first-audio latency p95 > 2.5 s for 15 min; tool error rate > 2%; call drop rate > 10%; any judge flag on disclosure or advice; cost per call > $0.60; stale scripts still taking calls (must be 0).
- **Rollback:** scripts are immutable versions; prompts and model configs live in versioned config; a per-trial feature flag pauses calls instantly.
- **Abuse and runaway cost:** signed, short-lived call tokens per trial; embed allowlist of approved domains; per-IP and per-phone daily limits; 12-minute cap enforced by the agent.
- **Data handling:** recordings and transcripts encrypted at rest, in a private store, with a retention period set by the sponsor; access logged.

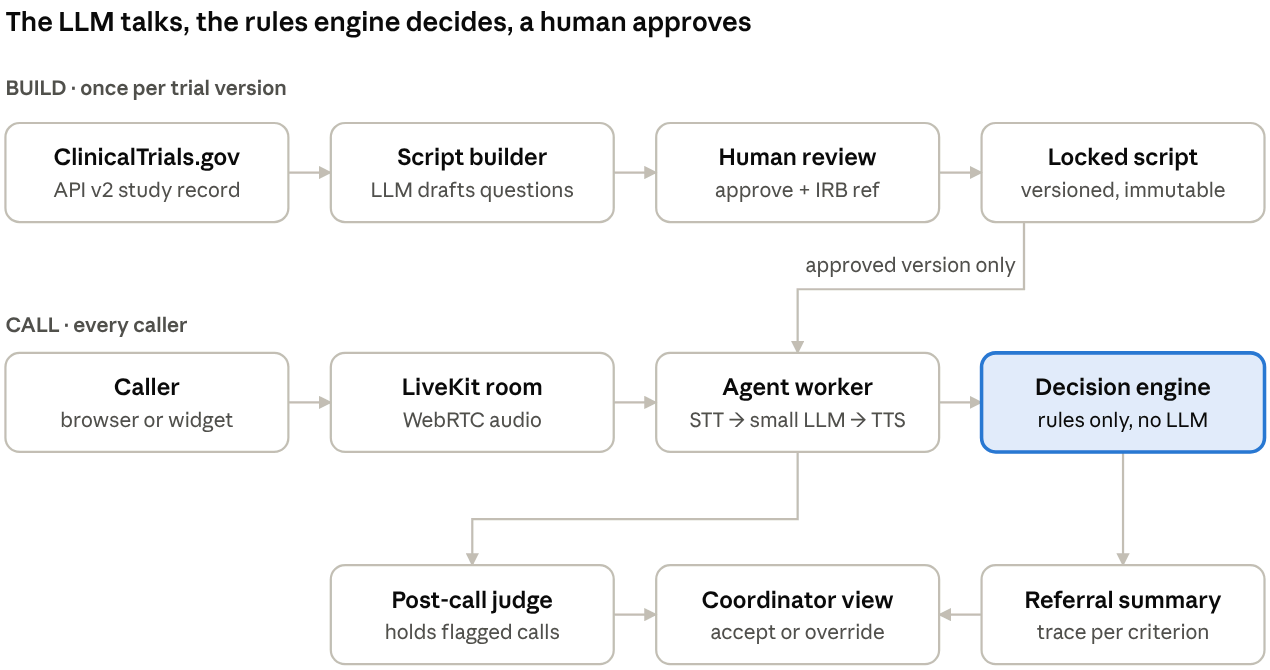
Launch checklist

- [ ] BAAs signed with all vendors in the audio and transcript path
- [ ] IRB approval of script, disclosure, consent and emergency message

- [ ] Offline eval gates met on the release candidate
- [ ] Security review: auth, token signing, embed allowlist, PHI storage
- [ ] Retention policy and deletion job in place
- [ ] Incident runbook and on-call owner named
- [ ] Coordinator trained on the dashboard and override flow

11. Engineering Context

Two paths share one database: a build path that turns a registry record into an approved script, and a call path that can only run that script.



Trial2Voice architecture · build path and call path

The decision engine is the only component that sets eligibility status; the LLM's job ends at structured answers.

- **Key integrations:** ClinicalTrials.gov API v2 (`GET /api/v2/studies/{nctId}`); LiveKit Cloud (rooms and agent hosting); Deepgram or AssemblyAI (STT); Anthropic and a small in-call model; Cartesia or ElevenLabs (TTS); email notification to coordinators.
- **Ownership:** Sankar owns product and build end to end; a part-time clinical reviewer owns the golden set and question review.
- **Dependencies not yet in place:** BAAs with voice vendors, approved emergency-message wording, a pilot IRB contact, and the 30-trial golden set.

12. Product Intelligence

The most valuable by-product is criterion-level drop-off: it shows the sponsor which criteria exclude the most interested callers.

- **Funnel tracked per trial:** link opened → call started → disclosure passed → consent given → all criteria answered → referral sent → site accepted → screening visit → enrolled (last two reported by the site).
- **Criterion drop-off:** share of callers excluded or marked Unsure at each criterion. Fed back to the sponsor as input for protocol amendments and future protocol design.
- **Feedback loop:** coordinator reject reasons and judge flags → proposed rewording of questions → new script version → re-approval. Nothing changes a live script without approval.
- **Personalization:** none across calls, by design. Within a call, only language and speech rate adapt.
- **Competitive context:** the alternatives are site call lines, recruitment call centers and web-form trial matchers. The wedge is voice, multilingual, after-hours, and a traced referral the site can trust.

13. Design & UX

The caller should know within ten seconds what this is, how long it takes, and that a person makes the final call.

- **Interaction model:** voice-first browser call with live captions and a progress bar ("Question 4 of 11"). Coordinators and sponsors use a web dashboard.
- **Opening (pre-rendered audio):** "Hi, I'm an AI assistant for the [study short name] study. I'll ask about 11 short questions to see if you might be a fit — about five minutes. I can't give medical advice, and the study team makes the final decision. This call is recorded. Is it OK to start?"
- **Latency handling:** fixed lines are pre-rendered; a short spoken filler only when a tool call exceeds 700 ms; captions appear as the caller speaks.
- **Uncertainty:** "I'm not sure" is always accepted ("That's fine — the study team can check that with you"). The closing line never says "eligible".
- **Edit and override:** the caller can say "go back" to change the last answer; the end of the call reads back key answers for confirmation. Coordinators can override any answer or status, with a required reason, logged.

- **Ending:** "Based on your answers, you may be a fit. The study team at [nearest site] will contact you within [N] business days." N is set by the site.
- **Accessibility:** captions, adjustable speech rate, and a text-input fallback for callers who can't speak.

14. Technical Implementation Framework (Next.js)

Next.js runs the web app and APIs; the real-time voice agent runs as a separate Python worker, because that is where the voice framework lives.

- **Frontend (App Router):**
  - `/admin/trials` and `/admin/trials/[nctId]/script` — ingest, script editor with criterion ↔ question view, version diff, approve. Server-rendered behind auth.
  - `/call/[trialSlug]` — public call page; client-rendered for WebRTC.
  - `/embed/[trialSlug]` — the same call UI inside an iframe for the widget.
  - `/coordinator` — referral queue, summaries, overrides.
- **API layer (route handlers):** `POST /api/trials` (ingest by NCT ID); `POST /api/trials/[id]/build` (streams build progress over SSE); `POST /api/scripts/[id]/approve`; `POST /api/call-token` (signed, short-lived room token per trial); `POST /api/voice/webhook` (room events).
- **Model integration:** script build and judge through the Anthropic SDK with JSON-schema tool output. In-call STT → LLM → TTS runs in the LiveKit Agents worker (Python), which calls the decision engine and writes answers to the database.
- **State and data:** Postgres (Supabase or Neon) via Drizzle for trials, script versions, approvals, calls, answers and referrals; Upstash Redis for rate limits; TanStack Query on the client.
- **Deployment:** Next.js on Vercel; agent worker on LiveKit Cloud agent hosting; secrets in platform environment variables; separate dev and prod projects. PHI-bearing services only on plans that offer a BAA.

15. Feature-Wise Build Plan

Eight weeks from Nov 2 to Dec 24, 2026, in four phases; each phase ends with something demoable.

Feature	Phase	Owner	Target date	Dependencies
NCT ID ingest + script builder (structured output)	1 · Script	Sankar	Nov 13	ClinicalTrials.gov API v2
Script editor, version diff, approval lock	1 · Script	Sankar	Nov 13	Script builder
Decision engine + unit tests on golden trials	1 · Script	Sankar + clinical reviewer	Nov 20	Golden set of 30 trials
Voice agent: disclosure, consent, question loop, read-backs	2 · Call	Sankar	Nov 27	LiveKit, STT, TTS accounts
Red-flag stop + fixed emergency message	2 · Call	Sankar	Dec 4	Approved message wording
Referral summary + coordinator dashboard	2 · Call	Sankar	Dec 4	Decision engine
Simulated-caller harness, judges, regression gate	3 · Evals	Sankar	Dec 11	Golden set, personas
English ↔ Spanish, nearest site, embed widget	3 · Evals	Sankar	Dec 18	Spanish STT/TTS voices
Monitoring, cost limits, protocol-change watch	4 · Ship	Sankar	Dec 24	All P0 features
Demo trials, walkthrough video, portfolio page	4 · Ship	Sankar	Dec 24	Eval gates met

The golden set and a clinical reviewer are the critical path: without them, Phase 3 gates can't be set.

Open Questions

1. Who is the pilot partner — a site, a CRO or a patient-advocacy group — and which IRB reviews the script?

2. Who is the data controller (sponsor, site or vendor)? That decides covered-entity vs. business-associate obligations.
3. Will the IRB accept an AI agent delivering the approved script with pre-approved alternate phrasings, or only verbatim?
4. Should callers who don't fit be offered other recruiting trials? Useful for patients, but a possible conflict for a single-sponsor deployment.
5. How long are recordings and transcripts kept, and may they be used to improve the product?
6. Which language after Spanish, and is it driven by trial geography or by the sponsor's diversity goals?

Assumptions Made

- Portfolio MVP built solo by Sankar in 8 weeks, with a part-time clinical reviewer for the golden set.
- The public demo uses real recruiting trials with a clear "demo — not affiliated with this study" banner and sends no real referrals.
- Only registry criteria are used; the full protocol is out of scope for the MVP.
- US-first; HIPAA and US IRB practice apply.
- Cost figures are estimates for a typical voice cascade at list prices; all targets are hypotheses for the pilot to test.

Sources

- Analysis of Rates of Completion, Delays, and Participant Recruitment in Randomized Clinical Trials in Surgery — JAMA Network Open, 2023
- Reasons for slow accrual in NCI phase 1 and 2 trials (table)
- Creighton University IRB — Requirements for Recruitment Materials (2025)
- UC San Diego OCR-005 — Research Participant Recruitment SOP
- FDA — Recruiting Study Subjects (guidance)
- ClinicalTrials.gov API v2 reference